

Estimating drought risk through classical and Bayesian modeling

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What is Drought?

Definitions

- ◀ Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow-onset disaster characterized by the lack of precipitation, resulting in a water shortage. (WHO)
- ◀ A drought is a period when an area or region experiences below-normal precipitation. The lack of adequate precipitation, either rain or snow, can cause reduced soil moisture or groundwater, diminished stream flow, crop damage, and a general water shortage. (National Geographic)

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Drought

- ◀ **DURATION:** A drought can last for months or years or may be declared in the early days if low rain perception is clear for some period.
- ◀ **IMPACTS:** Droughts can have a substantial impact on the ecosystem and agriculture of the affected region and harm the local population and economy.

Types of Drought



4 TYPES OF DROUGHT

AGRICULTURAL DROUGHT

Agricultural Droughts occur when there is not enough moisture in the soil to sustain the growth of crops.

SOCIOECONOMIC DROUGHT

Socioeconomic Droughts occur when the water supply is too low to support human and environmental needs

HYDROLOGICAL DROUGHT

Hydrological Droughts occur when there is a lack of surface and subsurface water supply.

METEOROLOGICAL DROUGHT

Meteorological Droughts are region-specific; they occur when an area receives less rainfall than it normally should.

Types of Drought

- ◀ **METEOROLOGICAL:**
Meteorological drought arises from insufficient precipitation, affecting water infiltration, runoff, and groundwater recharge.



Source: <https://www.shutterstock.com/image-photo/big-tree-on-drought-land-dry-1606270384>

Causes of Drought

◀ Human Causes

- Climate change
- Deforestation
- Agriculture
- High water demand

◀ Natural Causes

- Changes in ocean temperatures
- The jet stream



Source: <https://scijinks.gov/what-causes-a-drought/>

Drought in Baluchistan

◀ Historical Droughts

- Baluchistan province has faced many severe droughts in the past, i.e., 1967 to 1969, 1971, 1973 to 1975, 1994, and 1998 to 2002, which had a drastic impact on livelihoods and its economy. These prolonged droughts destroyed nearly 80% of fruit orchards and around two million animals died.
- These assessments and measures are according to the case studies by government, different environmentalist organization and United Nations Development Program (UNDP) case studies.
- The history shows that the drought effects on Baluchistan are mainly due to climate change happening around the world and most importantly happening in the Pakistan itself.

Motivation of the Study

- ◀ Safeguarding water and food security in Baluchistan.
- ◀ Protecting livelihoods from climate change impacts.
- ◀ Understanding and mitigating drought phenomena.
- ◀ Empowering decision-makers with knowledge and tools.
- ◀ Ensuring well-being of present and future generations.

Methodology

Study Area

- ◀ Weather station data sourced from the NASA website <https://power.larc.nasa.gov/data-access-viewer>.
- ◀ Data comprises daily aggregate precipitation (in millimeters) from 2 weather stations namely Musakhel and Ziarat from Baluchistan.
- ◀ Data covers 40 years from January 1st, 1984, to December 31st, 2023.
- ◀ The series of each is divided into two 20-year duration subseries: Period I (1984–2003) and Period II (2004–2023).

Extreme Value Theory

- ◀ Extreme value theory deals with rare events and comprises through block maxima and peak over threshold.
- ◀ Extreme values occurred at the tails of a probability distribution.
- ◀ Models: GEV and GPD.

Standardized Precipitation Index

- ◀ SPIs series is constructed for each period of both stations.
- ◀ Calculation based on the three-month sum of precipitation data for the corresponding period.

Table 1: Classification of drought levels, based on SPI values

SPI value	Drought category
≥ 2.00	Extremely humid
$1.50 \leq \text{SPI} \leq 2.00$	Very humid
$1.00 \leq \text{SPI} \leq 1.50$	Humid
$-1.00 \leq \text{SPI} \leq 1.00$	Normal
$-1.50 \leq \text{SPI} \leq -1.00$	Dry
$-2.00 \leq \text{SPI} \leq -1.50$	Very dry
$\text{SPI} \leq -2.00$	Extremely dry

Source: Labedzki, L. (2006). Susze rolnicze. zarys problematyki oraz metody monitorowania i klasyfikacji. Woda. Srodowisko. Obszary Wiejskie. Rozprawy Naukowe i Monografie, (17):1–107.

Classical Modeling

Generalized Extreme Value (GEV) distribution

- ▶ GEV distribution models the block maxima.
- ▶ The inverse CDF of GEV is written as

$$G_{\alpha,\beta,\kappa}(x) = \begin{cases} \exp \left\{ - \left[1 + \kappa \left(\frac{x-\alpha}{\beta} \right) \right]^{-\frac{1}{\kappa}} \right\} & \text{if } \kappa \neq 0, \kappa \left(\frac{x-\alpha}{\beta} \right) > 0 \\ \exp \left\{ \exp \left(\frac{x-\alpha}{\beta} \right) \right\} & \text{if } \kappa = 0, x \in \mathbb{R}. \end{cases} \quad (1)$$

- ▶ The log-likelihood function for unknown parameters vector $\theta = (\alpha, \beta, \kappa)$ is defined as

$$l(\theta) = -n \log \beta - \left(1 + \frac{1}{\kappa} \right) \sum_{i=1}^n \log \left[1 + \kappa \frac{(x_i - \alpha)}{\beta} \right] - \sum_{i=1}^n \left[1 + \kappa \frac{(x_i - \alpha)}{\beta} \right]^{-\frac{1}{\kappa}} \quad (2)$$

Bayesian Modeling

- ▶ The GEV distribution parameters $\boldsymbol{\theta} = (\alpha, \beta, \kappa)$ are considered random variables for which we identify the prior distributions.
- ▶ $\boldsymbol{\theta}$ in terms of posterior PDF, is given by

$$\pi(\boldsymbol{\theta}|x) = \frac{L(x|\boldsymbol{\theta})g_{\boldsymbol{\theta}}(\boldsymbol{\theta})}{\int_{\Theta} L(x|\boldsymbol{\theta})g_{\boldsymbol{\theta}}(\boldsymbol{\theta})d\boldsymbol{\theta}}, \quad (3)$$

- ▶ The joint distribution for $\boldsymbol{\theta}$ was supposed to be written as

$$g_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = g_{\boldsymbol{\theta}}(\alpha, \beta^* = \log(\beta), \kappa) = g_{\alpha}(\alpha)g_{\beta^*}(\beta^*)g_{\kappa}(\kappa). \quad (4)$$

- ▶ For marginals, we consider non-informative priors as

$$g_{\alpha}(\alpha) \sim \mathcal{N}(0, 10000) \quad g_{\beta^*}(\beta^*) \sim \mathcal{N}(0, 10000) \quad g_{\kappa}(\kappa) \sim \mathcal{N}(0, 100). \quad (5)$$

Risk Assessment

- ◀ Drought risk is the likelihood of experiencing a drought.
- ◀ The risk is defined as

$$R_{\mathcal{D}}(G_{\alpha,\beta,\kappa}, d_{cr}) = \mathcal{D} \quad (6)$$

and

$$R_{\mathcal{D}_Q}(G_{\alpha,\beta,\kappa}^{-1}, p_{cr}) = \mathcal{D}_Q, \quad (7)$$

Source: Piwowar, A. and Ku zmi nski, L. (2023). Drought risk probabilistic models based on extreme value theory. Environmental Science and Pollution Research, 30(42):95945–95958.

Estimated Parameters

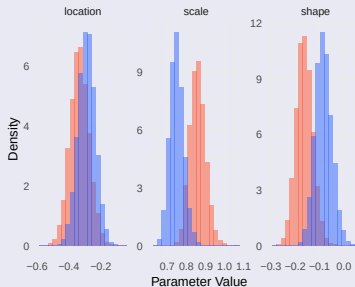
Table 2: Estimator values for parameters of distribution functions $G_{\mu,\sigma,\kappa}$ along with compatibility test results

Site	Method	Period	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$
Musakhel	MLE	I	-0.332 (0.060)	0.852 (0.041)	-0.166 (0.035)
		II	-0.289 (0.052)	0.742 (0.036)	-0.087 (0.034)
	Bayesian	I	-0.335 (0.061)	0.860 (0.042)	-0.162 (0.035)
		II	-0.290 (0.053)	0.750 (0.037)	-0.084 (0.035)
Ziarat	MLE	I	-0.299 (0.066)	0.927 (0.045)	-0.252 (0.035)
		II	-0.284 (0.060)	0.852 (0.040)	-0.186 (0.033)
	Bayesian	I	-0.303 (0.065)	0.934 (0.045)	-0.245 (0.035)
		II	-0.287 (0.061)	0.860 (0.041)	-0.183 (0.033)

Posterior Densities

Posterior densities of GEV parameters

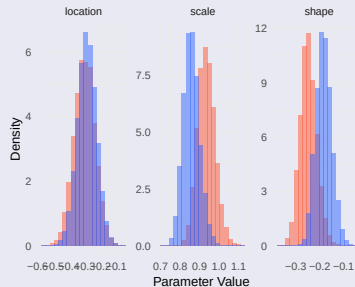
Musakhel Period I Period II



Posterior Densities for
Musakhel

Posterior densities of GEV parameters

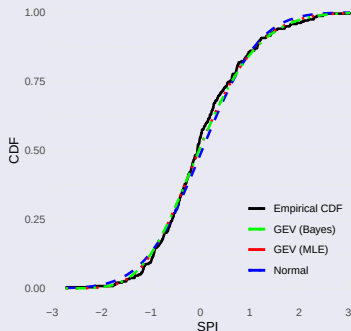
Ziarat Period I Period II



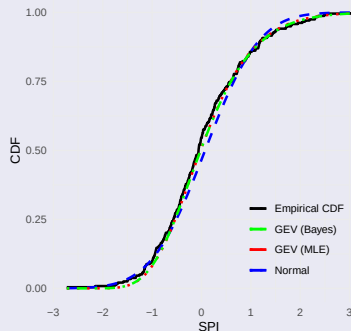
Posterior Densities for
Ziarat

CDF Plots for Musakhel

CDF plot (Musakhel Period I)

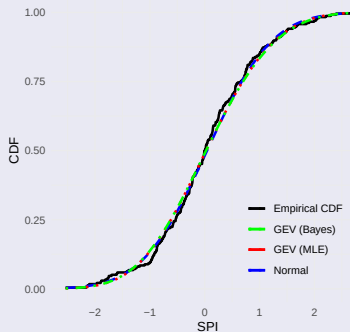


CDF plot (Musakhel Period II)

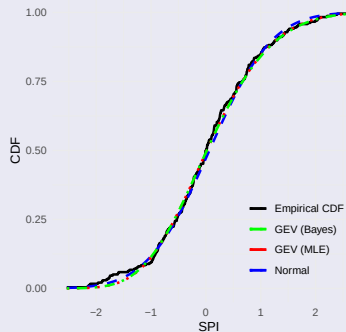


CDF Plots for Ziarat

CDF plot (Ziarat period I)



CDF plot (Ziarat period II)



Drought Risk Measures

Table 3: Measures of drought risk for Musakhel and Ziarat in periods I and II, estimated with the probabilistic model of drought risk.

Site	Method	Period	$R_{\mathcal{D}}(G_{\alpha,\beta,\kappa}, d_{cr}) = \mathcal{D}$
Musakhel	MLE	I	0.008029
		II	0.000744
	Bayesian	I	0.008353
		II	0.000787
Ziarat	MLE	I	0.010965
		II	0.003971
	Bayesian	I	0.011214
		II	0.004169
Color indication:	Very high risk		$\mathcal{D} \geq 0.05$
	High risk		$0.03 \leq \mathcal{D} < 0.05$
	Moderate risk		$0.01 \leq \mathcal{D} < 0.03$
	Low risk		$0.001 \leq \mathcal{D} < 0.01$
	Insignificant risk		$\mathcal{D} < 0.001$

Source for scale: Piwowar, A. and Ku zmi nski, L. (2023). Drought risk probabilistic models based on extreme value theory. Environmental Science and Pollution Research, 30(42):95945–95958.

Conclusion

Conclusion

- ◀ Despite extreme observations in the data, the proposed framework is flexible and efficient for measuring drought.
- ◀ In both regions a declining trend in drought is observed.
- ◀ The results declare that Musakhel is the safest region for living as compared to Ziarat.

Future Work

Work in progress

- ◀ Many stations are under observation.
- ◀ Use of informative priors to improve the results.
- ◀ Construction of priors through historical records of other neighboring stations (Ahmad et al., 2022)^a.

^aAhmad, T., Ahmad, I., Arshad, I. A., Bianco, N. (2022). A comprehensive study on the Bayesian modelling of extreme rainfall: A case study from Pakistan. *International Journal of Climatology*, **42**(1), 208-224.

THANK YOU

Any Questions?